

Minghao Guo

PERSONAL INFORMATION

Name:  Minghao Guo (郭明浩)
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Princeton, NJ 08544, USA
GitHub:  [mh-guo](https://github.com/mh-guo)
GitLab:  [mhguo](https://gitlab.com/mhguo)

POSITION

Junior Fellow, Simons Society of Fellows

Columbia University

- Host: Prof. Greg Bryan

New York City, NY, USA

2026 – Expected 2029

EDUCATION

Princeton University

Graduate Student, Department of Astrophysical Sciences

- Advisors: Eliot Quataert, James M. Stone
- Thesis: Black Hole Accretion and Feedback from Galaxies to Event Horizons

Princeton, NJ, USA

2021 – 2026

Princeton University

Master of Arts in Astrophysical Sciences, Department of Astrophysical Sciences

Princeton, NJ, USA

2021 – 2023

Peking University

Bachelor of Science in Physics, Yuanpei College

- Advisors: Luis C. Ho, Kohei Inayoshi, Lijing Shao
- Thesis: A Numerical Study of Scalar-tensor Gravity Theory

Beijing, China

2016 – 2021

EXPERIENCE

Institute for Advanced Study

Visiting Graduate Student, School of Natural Sciences

Princeton, USA

Dec. 2023 – 2026

RESEARCH INTERESTS

- Black holes, high energy astrophysics, accretion disk, general relativistic magnetohydrodynamics (GRMHD)
- Active galactic nuclei (AGN), galaxy formation and evolution, multiphase interstellar medium (ISM)
- Numerical methods and simulations, GPU computing, new numerical techniques
- Neutron stars, pulsars, gravitational waves, modified gravity theory

PUBLICATIONS

Full list: [ADS](#), [Google Scholar](#), or [ORCID](#). Citations (ADS): > 200 first-author; > 300 total.

1. **Minghao Guo**, James M. Stone, Eliot Quataert, and Volker Springel, “Cyclic Zoom: Multiscale GRMHD Modeling of Black Hole Accretion and Feedback,” [ApJ](#) **987**, 202 (2025), [arXiv:2504.16802 \[astro-ph.HE\]](#) .
2. **Minghao Guo**, Eliot Quataert, Jonathan Squire, Philip F. Hopkins, and James M. Stone, “Idealized Global Models of Accretion Disks with Strong Toroidal Magnetic Fields,” arXiv e-prints, arXiv:2505.12671 (2025), [arXiv:2505.12671 \[astro-ph.HE\]](#) .
3. **Minghao Guo**, Chang-Goo Kim, and James M. Stone, “Evolution of Supernova Remnants in a Cloudy Multiphase Interstellar Medium,” [ApJ](#) **990**, 49 (2025), [arXiv:2411.12809 \[astro-ph.GA\]](#) .

4. **Minghao Guo**, James M. Stone, Eliot Quataert, and Chang-Goo Kim, “Magnetized Accretion onto and Feedback from Supermassive Black Holes in Elliptical Galaxies,” [ApJ **973**, 141 \(2024\)](#), highlighted in [@PlotAstro](#), [arXiv:2405.11711 \[astro-ph.HE\]](#) .
5. **Minghao Guo**, James M. Stone, Chang-Goo Kim, and Eliot Quataert, “Toward Horizon-scale Accretion onto Supermassive Black Holes in Elliptical Galaxies,” [ApJ **946**, 26 \(2023\)](#), highlighted in [AAS Nova](#), [arXiv:2211.05131 \[astro-ph.HE\]](#) .
6. **Minghao Guo**, Junjie Zhao, and Lijing Shao, “Extended reduced-order surrogate models for scalar-tensor gravity in the strong field and applications to binary pulsars and gravitational waves,” [PhRvD **104**, 104065 \(2021\)](#), [arXiv:2106.01622 \[gr-qc\]](#) .
7. **Minghao Guo**, Kohei Inayoshi, Tomonari Michiyama, and Luis C. Ho, “Hunting for Wandering Massive Black Holes,” [ApJ **901**, 39 \(2020\)](#), [arXiv:2006.08203 \[astro-ph.HE\]](#) .
8. **Minghao Guo**, Min Du, Luis C. Ho, Victor P. Debattista, and Dongyao Zhao, “A New Channel of Bulge Formation via the Destruction of Short Bars,” [ApJ **888**, 65 \(2020\)](#), [arXiv:1911.07002 \[astro-ph.GA\]](#) .
9. Hai-Yang Wang, **Minghao Guo**, Elias R. Most, Philip F. Hopkins, and Aretaios Lalakos, “Galactic-scale Feeding Reveals Warped Hypermagnetized Multiphase Circumbinary Accretion Around Supermassive Black Hole Binaries,” [arXiv e-prints](#) , [arXiv:2504.03874 \(2025\)](#), help with numerical tools, design simulations, data analysis, paper writing, [arXiv:2504.03874 \[astro-ph.HE\]](#) .
10. Rebecca Diesing, **Minghao Guo**, Chang-Goo Kim, James Stone, and Damiano Caprioli, “Nonthermal Signatures of Radiative Supernova Remnants,” [ApJ **974**, 201 \(2024\)](#), provide simulations results, discussion, [arXiv:2404.15396 \[astro-ph.HE\]](#) .
11. Julie Hlavacek-Larrondo, Hyunseop Choi, **Minghao Guo**, Annabelle Richard-Laferrrière, Carter Rhea, Marine Prunier, Helen Russell, Andy Fabian, Jonelle L. Walsh, Marie-Joëlle Gingras, Brian McNamara, Steve Allen, André-Nicolas Chené, Alastair Edge, Marie-Lou Gendron-Marsolais, Michael McDonald, Priyamvada Natarajan, Jeremy Sanders, James F. Steiner, Benjamin Vigneron, and Anja von der Linden, “Hubble Space Telescope Observations within the Sphere of Influence of the Powerful Supermassive Black Hole in PKS 0745-191,” [ApJ **980**, 170 \(2025\)](#), provide theoretical predictions and comparisons, paper writing, [arXiv:2501.03339 \[astro-ph.GA\]](#) .
12. James M. Stone, Patrick D. Mullen, Drummond Fielding, Philipp Grete, **Minghao Guo**, Philipp Kempfski, Elias R. Most, Christopher J. White, and George N. Wong, “AthenaK: A Performance-portable Version of the Athena++ Adaptive Mesh Refinement Framework,” [ApJS **283**, 27 \(2026\)](#), [arXiv:2409.16053 \[astro-ph.IM\]](#) .
13. Rajsekhar Mohapatra, Eliot Quataert, Drummond Fielding, and **Minghao Guo**, “The Type Ia Supernova and Asymptotic Giant Branch Stellar Ejecta-regulated Interstellar Medium of Massive Galaxies,” [ApJ **989**, 103 \(2025\)](#), provide tools and discussion, [arXiv:2502.05329 \[astro-ph.GA\]](#) .

APPROVED PROPOSALS

INCITE 2026 AWARD (Co-PI)	~ 4 M GPU hours
Exploring the Frontiers of Black Hole Accretion and Feedback	Jan. 2026 – Jan. 2027
NSF ACCESS Maximize (Co-PI)	~ 100 k GPU hours
Investigating stellar and black hole heating in massive galaxies, groups and clusters	Oct. 2025 – Sep. 2026
EuroHPC (Co-PI)	~ 1 M GPU hours
Multi-scale (GR)MHD modelling of accretion onto supermassive black holes	Mar. 2024 – Jun. 2025
NSF ACCESS Maximize (Co-PI)	~ 100 k GPU hours
Investigating stellar and black hole heating in massive galaxies, groups and clusters	Oct. 2024 – Sep. 2025

JWST Cycle 3 (Co-PI)	~ 6 hours
Mapping a Black Hole Accretion Flow with JWST/NIRSpec	2024 – 2025
NSF ACCESS Accelerate (Co-PI)	~ 42 k GPU hours
Multiscale GRMHD Modeling of Accretion onto Supermassive Black Holes	Nov. 2023 – Feb. 2025
NSF ACCESS Explore (PI)	~ 10 k GPU hours
Multi-scale MHD Modeling of Accretion onto Supermassive Black Holes	Jun. 2023 – Jun. 2024

REFERENCES

Charles A. Young Professor of Astronomy, Eliot Quataert	Princeton University
quataert@princeton.edu	
Professor James M. Stone	Institute for Advanced Study
jmstone@ias.edu	
Prof. Dr. Volker Springel	Max Planck Institute for Astrophysics
vspringel@mpa-garching.mpg.de	
Ira S. Bowen Professor, Philip F. Hopkins	California Institute of Technology
phopkins@caltech.edu	
Director, Chair Professor Luis C. Ho	Peking University
lho.pku@gmail.com	
Professor Kohei Inayoshi	Peking University
inayoshi.pku@gmail.com	
Professor Lijing Shao	Peking University
lishao@pku.edu.cn	

HONORS AND AWARDS

Princeton First Year Fellowship in Natural Science & Engineering	2021
Weiming Bachelor, Peking University	Jun. 2021
Yuanpei Outstanding Young Scholars	Dec 2020
Lin-bridge First Prize for Undergraduate Research	Sep. 2020
Yuanpei College First Award for Undergraduate Research	Jun. 2020
Xingcheng Award for Undergraduate Research	May 2019
National Undergraduate Research & Training Program	May 2019
Peking University Scholarship for Outstanding Freshmen	Sep. 2016

TEACHING AND MENTORING

Co-advising Princeton Undergraduate: Milo Salvucci	2025
Graduate Student Peer Mentor	2025
Teaching Assistant for Cosmology	Spring 2024
Teaching Assistant for General Relativity	Fall 2023
Co-advising Princeton Undergraduate: Sajia Shahrin Neha	2023
Lecture on Flatiron Institute CCA Fluid Dynamics Summer School	Aug. 2023
<i>GPU Computing using AthenaK: Black Hole Accretion and Supernova Remnants</i>	

SERVICE AND OUTREACH

Reviewer for the Astrophysical Journal (ApJ), Open Journal of Astrophysics, European Physical Journal

[Prison Teaching Initiative](#)
Teaching Astronomy

Jan. 2025 - Now

Member of the Learning the Universe Collaboration

Sep. 2022 – Now

[Astronomy on Tap Trenton: How does a tiny black hole affect the entire galaxy?](#)

Mar. 2025

Member of the Local Organizing Committee, Learning the Universe Collaboration meeting

Mar. 2025

Member of the Mental Health Working Group

2023 – 2024

Co-organizer of the Princeton Astrophysics Thunch, Astronomy

2022 – 2023

TECHNICAL SKILLS

Programming: Proficient in Python, C/C++, L^AT_EX, Mathematica, Git; Basic knowledge of Matlab, Fortran, and HTML/CSS.

Software and Packages: AthenaK, Athena++, MPI, OMP, cuda, SymPy, yt, emcee, VisIt, ParaView, PLUTO, IRAF, GALFIT

Techniques: Numerical simulations, massive parallel computing on supercomputer, dataset analyzing and visualization.

SOFTWARE DEVELOPMENT

- [AthenaK](#) (Developer): Block-based AMR framework with fluid, particle and numerical relativity solvers in Kokkos.
- [AthenaKit](#) (Owner): Toolkit for analysis and visualization of simulations by AthenaK
- [pySTGROMX](#) (Owner): Extended reduced-order surrogate models for scalar-tensor gravity of Damour and Esposito-Farèse (DEF)

PRESENTATIONS AND CONFERENCES

Harvard BHI Workshop: Bridging Scales in the Black Hole Accretion-Feedback Problem Boston, MA
(Invited talk) *Accretion and Feedback from Galaxies to Horizons* May 2026

FCAD Astronomy Seminar Amherst, MA
(Invited talk) *Accretion and Feedback from Galactic to Horizon Scales* Apr. 2026

Leicester Astrophysics Seminar Leicester, England
(Oral talk) *Accretion and Feedback from Galactic to Horizon Scales* Jan. 2026

Caltech TAPIR Seminar Pasadena, CA
(Oral talk) *Accretion and Feedback from Galaxies to Event Horizons* Oct. 2025

UC Santa Barbara Astro Lunch Santa Barbara, CA
(Oral talk) *Accretion and Feedback from Galaxies to Event Horizons* Oct. 2025

KIPAC Compact Objects Group meeting Stanford, CA
(Oral talk) *Multiscale structure of black hole accretion flows* Oct. 2025

KIPAC Tea Talk Stanford, CA
(Oral talk) *Accretion and Feedback from Galaxies to Event Horizons* Oct. 2025

Theoretical Astrophysics Center Seminar Berkeley, CA
(Oral talk) *Accretion and Feedback from Galaxies to Event Horizons* Oct. 2025

Flatiron Institute CCA Numerical Series (Invited talk) <i>Cyclic Zoom: Multiscale GRMHD modeling of Black Hole Accretion and Feedback</i>	New York Jun. 2025
Northwestern CIERA Theory Group Meeting (Oral talk) <i>Accretion and Feedback from Galactic to Horizon Scales</i>	Evanston, Illinois Jun. 2025
Flatiron Institute CCA Coffee Talk (Oral talk) <i>Accretion and Feedback from Galactic to Horizon Scales</i>	New York May 2025
Caltech TAPIR Astronomy Talk (Oral talk) <i>Evolution of Supernova Remnants in a Cloudy Multiphase Interstellar Medium</i>	Pasadena, CA May 2025
The Institute for Theory and Computation (ITC) luncheons (Oral talk) <i>Magnetized Accretion onto and Feedback from Supermassive Black Holes</i>	Boston, MA Apr. 2025
UMich extreme-astroph seminar (Oral talk) <i>Multi-scale GRMHD Modeling of Black Hole Accretion and Feedback over Galactic Scales</i>	Ann Arbor, Michigan Apr. 2025
The 245th Meeting of the American Astronomical Society (Oral talk) <i>Magnetized Accretion onto and Feedback from Supermassive Black Holes</i>	National Harbor, Maryland Jan. 2025
DCC Workshop: Deciphering the Cosmic Code for Galaxy Formation (Oral talk) <i>Magnetized Accretion onto and Feedback from Supermassive Black Holes</i>	Puerto Varas, Chile Dec. 2024
Harvard BHI Workshop: Bridging Scales in the Black Hole Accretion-Feedback Problem (Invited talk) <i>Multi-Scale Simulations of Galaxy-SMBH Feeding</i>	Boston, MA May 2024
UC Santa Barbara Astro Lunch (Oral talk) <i>Evolution of Supernova Remnants in a Cloudy Multiphase Interstellar Medium</i>	Santa Barbara, CA Jan. 2024
KITP Program: Turbulence in Astrophysical Environments (Oral talk) <i>Toward Horizon-scale Accretion onto Supermassive Black Holes in Elliptical Galaxies</i>	Santa Barbara, CA Jan. 2024
Black Holes on Broadway: The Next Generation of AGN Models in Galaxy Formation (Oral talk) <i>Toward Horizon-scale Accretion onto Supermassive Black Holes in Elliptical Galaxies</i>	New York Dec. 2023
PKU KIAA Seminar (Oral talk) <i>Toward Horizon-scale Accretion onto Supermassive Black Holes in Elliptical Galaxies</i>	Beijing, China Oct. 2023
Galaxy Formation in Hangzhou: Observations and Physics of AGN Feedback (Oral talk) <i>Toward Horizon-scale Accretion onto Supermassive Black Holes in Elliptical Galaxies</i>	Hangzhou, China Oct. 2023
The Second Athena++ Workshop (Oral talk) <i>Toward Horizon-scale Accretion onto Supermassive Black Holes in Elliptical Galaxies</i>	New York May 2023
Learning the Universe Annual Meeting (Oral talk) <i>Accretion of Supermassive Black Holes in Elliptical Galaxies</i>	New York Sep. 2022
The 240th Meeting of the American Astronomical Society (Poster presentation) <i>Accretion of Supermassive Black Holes in Elliptical Galaxies</i>	Pasadena, CA Jun. 2022
2020 PKU-DoA Undergraduate Astronomy Symposium (Oral talk) <i>Hunting for Wandering Massive Black Holes</i>	Beijing, China Sep. 2020
2019 PKU-DoA Undergraduate Astronomy Symposium (Oral talk) <i>A New Channel of Bulge Formation via the Destruction of Short Bars</i>	Beijing, China Sep. 2019

2019 Annual Meeting of Chinese Astronomical Society

Delingha, China

(Oral talk) *A New Channel of Bulge Formation via the Destruction of Short Bars*

Sep. 2019

São Paulo School of Advanced Science on First Light

São Paulo, Brazil

Aug. 2019

IAU Symposium 353: Galactic Dynamics in the Era of Large Surveys

Shanghai, China

(Poster) *The Role of Short Bar Destruction in Regulating the Co-evolution of Black Holes and Bulges* Jun. 2019